

Class05

Alishba

TITLE HERE

Subtitle

Subsubtitle

stuff in bold

not in bold

indent text

something italics

Q1: Make a ggplot plot

Answer Q1: gherferfer

```
x <- 1:5  
x + 1
```

```
[1] 2 3 4 5 6
```

Let's make a graph

```
library()
```

Fixing Bad code:

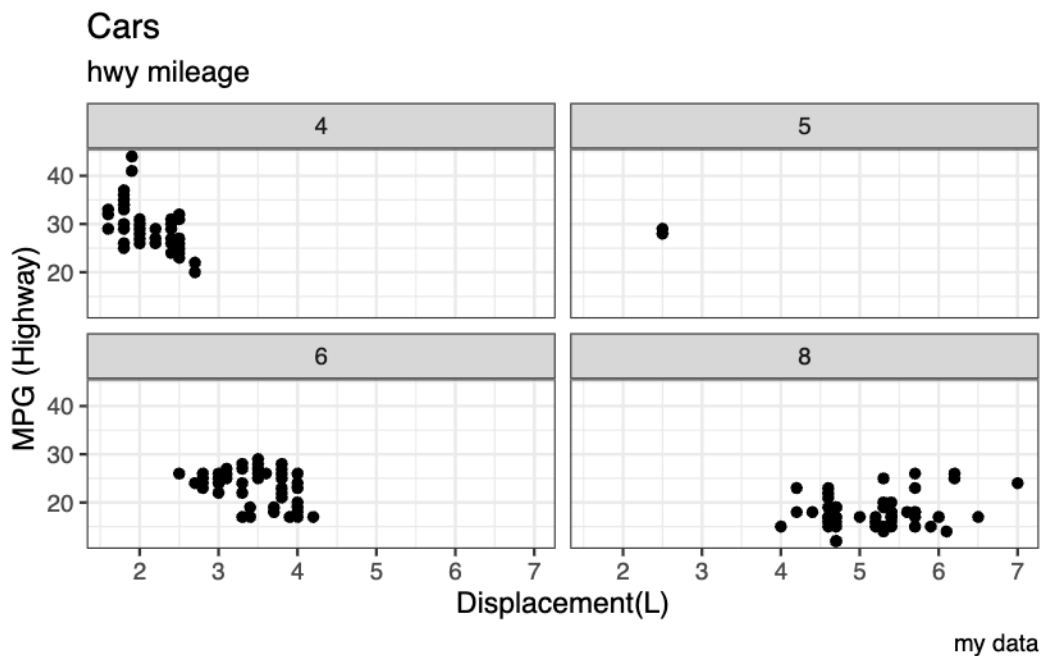
```
4 + 2
```

[1] 6

Plots:

```
# loaded ggplot2 package
library(ggplot2)

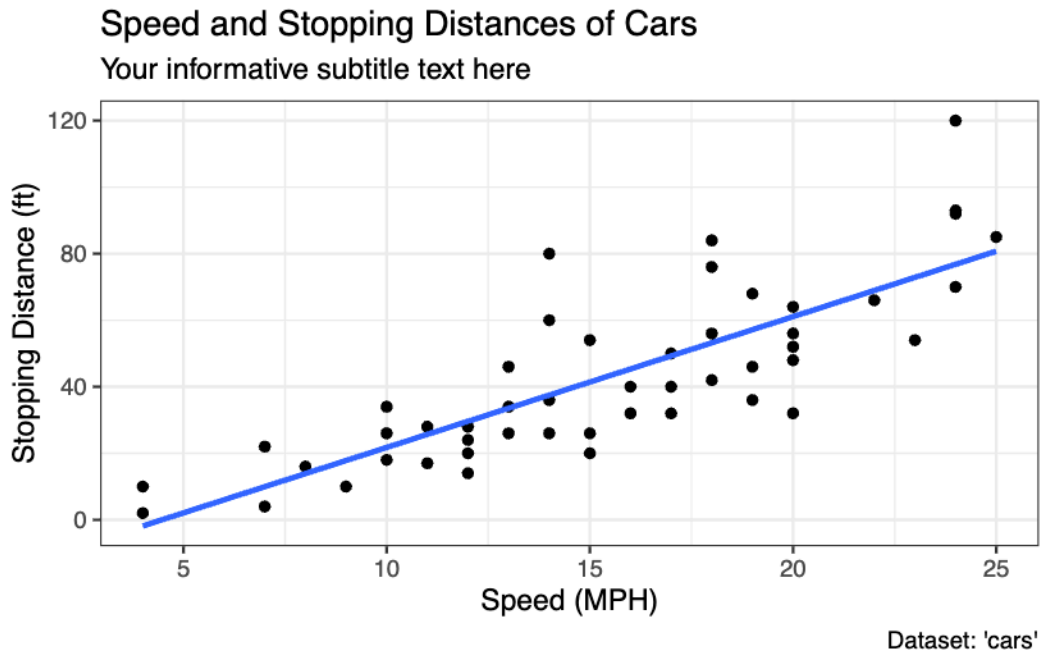
# specified data set, then determined the aesthetics
ggplot(mpg) +
  aes(x=displ, y=hwy) +
  # visualized data using scatterplot; included trendline
  geom_point() +
  theme_bw() +
  labs(title="Cars", subtitle="hwy mileage", caption="my data", x="Displacement(L)", y="MPG")
  facet_wrap(~cyl)
```



```
# made another graph using different data set, aesthetics, etc.
library(ggplot2)
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  labs(title="Speed and Stopping Distances of Cars", x="Speed (MPH)", y="Stopping Distance (ft)")
```

```
geom_smooth(method="lm", se=FALSE) +  
theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'



Q: Which geometric layer should be used to create scatter plots in ggplot2?

A: `geom_point()`

In another data set, I created a plot answered questions about gene expression.

```
# created a new data set on anti-viral drugs  
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"  
genes <- read.delim(url)  
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging

```

5      AATK  0.4711421  0.5598642  unchanging
6 AB015752.4 -3.6808610 -3.5921390  unchanging

```

```

# found number of genes in data set
nrow(genes)

```

```
[1] 5196
```

```

#found column names and number of columns
colnames(genes)

```

```
[1] "Gene"      "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

```

# determined number of 'up' regulated genes
table(genes$State)

```

```

      down unchanging      up
      72      4997      127

```

```

#determined fraction of up-regulated genes, rounded it
round( table(genes$State)/nrow(genes) * 100, 2 )

```

```

      down unchanging      up
      1.39      96.17      2.44

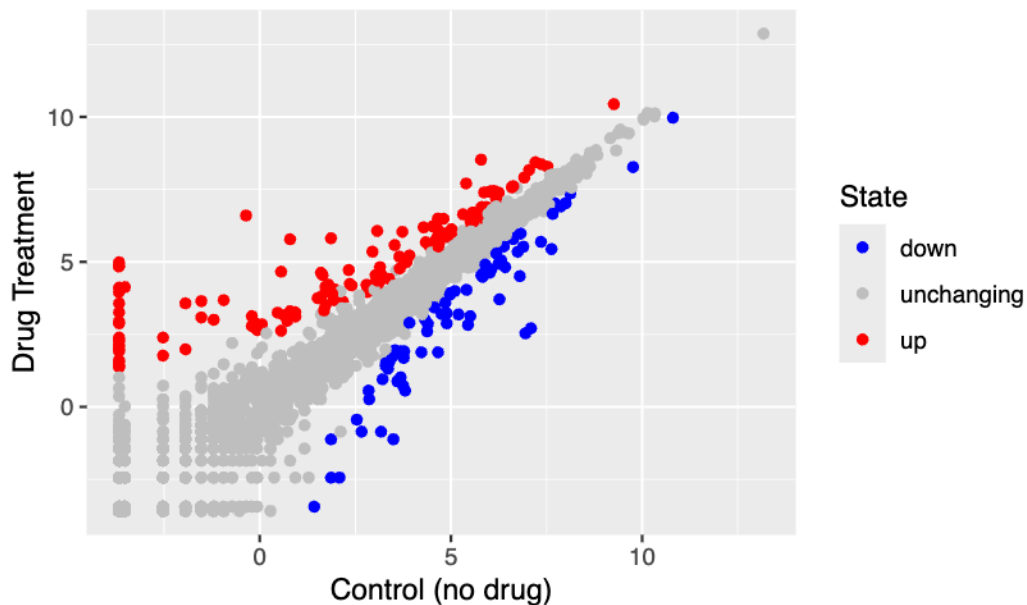
```

```

# created basic scatter plot of data set
p <- ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
p + scale_colour_manual( values=c("blue", "gray", "red")) +
  labs(title="Gene Expression Changes Upon Drug Treatment", x="Control (no drug)", y="Drug Treatment")

```

Gene Expression Changes Upon Drug Treatment



Q: Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

Answer: 5196 genes.

Q: Use the `colnames()` function and the `ncol()` function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

Answer: 4 columns.

Q: Use the `table()` function on the `State` column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

Answer: 127 'up' regulated genes.

Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

Answer: 2.44% of total genes are up-regulated.